



香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

DDA2001: Introduction to Data Science

Introduction

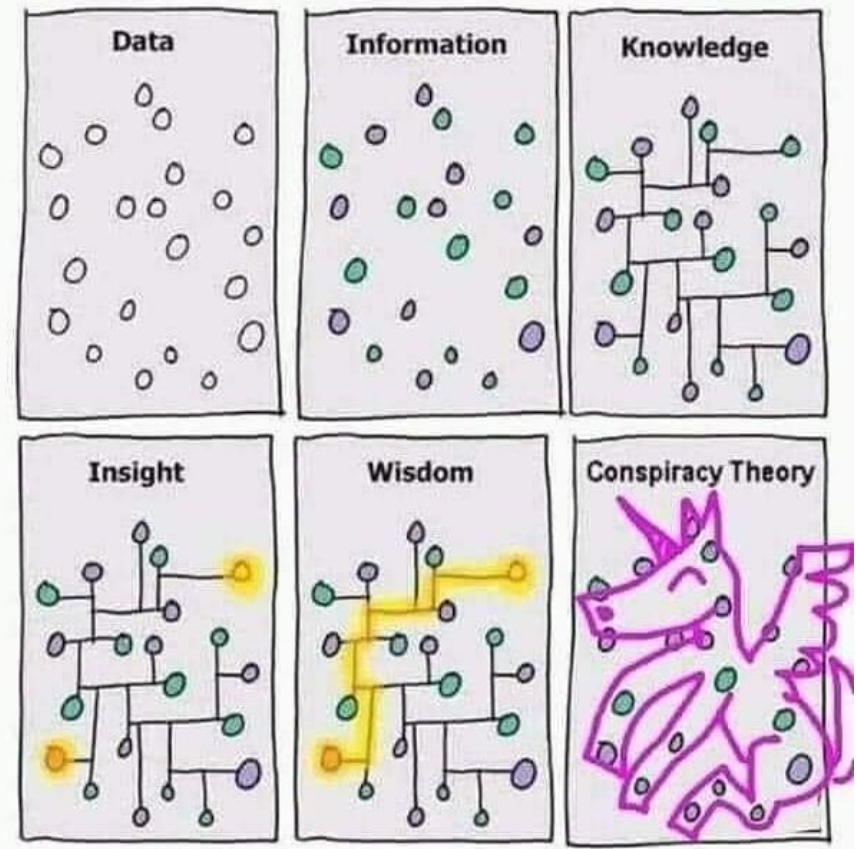
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Review of the Last Lecture

What is data science?

- Data science is the domain of study that deals with vast volumes of data using modern tools and techniques to **find unseen patterns**, **derive meaningful information**, and **make decisions**.



Use Data Science Tools to Solve Real Problems

- As a merchant, how do you price or choose what products to recommend to the customer?



Use Data Science Tools to Solve Real Problems

- **Probability** allows us to model the **likelihood** of different events, such as:
 - **How do customers choose from a set of products?**
 - **What factors influence their purchasing decision**
- Example: The **Multinomial Logit Model (MNL)** helps describe how customers choose products. The choice probability is:

$$P_i(S) = \frac{a_i}{1 + \sum_{j \in S} a_j}$$

where:

- a_i represents the attractiveness/utility/preference score of product i
- S is the set of all products or a product set offered
- The likelihood of choosing a product is proportional to its utility

Understanding $P_i(S) = \frac{a_i}{1 + \sum_{j \in S} a_j}$

Probabilities Depend Only on Ratios:

- The choice probability $P_i(S)$ is:

$$P_i(S) = \frac{a_i}{1 + \sum_{j \in S} a_j}$$

where

- The probability is a ratio of a_i to the total utility (1 as the baseline + sum of all a_j values).
- Since the **probabilities depend only on relative utilities**, the baseline utility doesn't need to be estimated explicitly.

Understanding $P_i(S) = \frac{a_i}{1 + \sum_{j \in S} a_j}$

Probability of Not Choosing Any Product:

$$1 - \sum_{i \in S} P_i(S) = \frac{1}{1 + \sum_{j \in S} a_j}$$

This probability is automatically derived from the sum of a_i values, without requiring us to explicitly learn the baseline utility.

Understanding $P_i(S) = \frac{a_i}{1 + \sum_{j \in S} a_j}$

The Baseline Utility Is Arbitrary:

- The baseline utility is fixed to 1 for simplicity, but it could be any constant c .
- If c were used instead of 1, all a_i values would scale proportionally, and the resulting probabilities would remain unchanged.

This means:

- The **absolute value** of a_i is not important—only its value **relative to the other** a_j matters.

Understanding $P_i(S) = \frac{a_i}{1 + \sum_{j \in S} a_j}$

Simplification in Learning:

- By fixing the baseline utility to 1, we avoid the need to estimate it from the data.
- All the information about choice probabilities is encoded in the a_i values, which can be directly learned from the data.

The goal of introducing the **Multinomial Logit Model** here is not to explain the model itself, but to show how such tools can quantify **choice probabilities** by capturing the relative likelihood of decisions based on utility.

It highlights how we can use such probabilistic models to analyze and predict real-world decision-making.

Use Data Science Tools to Solve Real Problems

- **Statistics** allows you to:
 - **Estimate parameters:** Determine the values of parameters like a_i in the multinomial logit model.
 - **Validate the model:** Ensure that the probability model actually reflects real-world consumer behavior.
- **Machine Learning:**
If the simple model (like the multinomial logit model) isn't enough, machine learning can help **build and learn complex models**.
- For example:
- $P_i(S)$ may depend on more complex features like:
 - Product attributes (e.g., color, price, brand)
 - Consumer profiles (e.g., age, location, past behavior)

Use Data Science Tools to Solve Real Problems

- **Optimization**

After building a model and predicting how customers will behave, you want to **maximize** your objectives—such as maximizing revenue or sales.

- For example, you want to choose the products to offer that will maximize your revenue:

$$\max_S \sum_{i \in S} r_i P_i(S)$$

where:

- Suppose the choice rule $P_i(S)$ is given.
- Let r_i be the revenue of product i .
- Choose a set of products to maximize the above objective.

Goal of Today's Lecture

We will provide more examples to deepen your understanding of when to use probability, statistics, machine learning, and optimization tools to solve real-world data science problems.

- What are these tools, and
- What can they help you achieve?

Example 1



Blind Box Example

- Ten of you went to a blind box seller for the following blind box. You were told the chance to get a Harry Potter versus a Lord Voldemort is fifty-fifty.
- You all wish to get a Harry Potter.



Blind Box Example

- You each bought one blind box. When you opened it, you found...



- What do you think? *Do you believe the seller told the truth about the 50-50 distribution?*

The Power of Probability – Can It Happen?

- If the seller was truthful, the probability of getting all Voldemorts is incredibly small !!

The Power of Probability – Can It Happen?

- If the seller was truthful, the probability of getting all Voldemorts is incredibly small !!
- How likely is This? Let's compute:

$$P(\text{all voldemort}) = 0.5^{10} \approx 0.0009765625$$

This means there is less than a 0.1% chance of this happening if the seller's claim is true

Blind Box Example

- If the seller told the truth, the probability that all ten of you got a Voldemort is < 0.001 . —→ Probability
- Given the above observation, you are pretty confident that the seller was lying. What's the true chance of getting a Harry Potter versus Voldemort? —→ Statistics

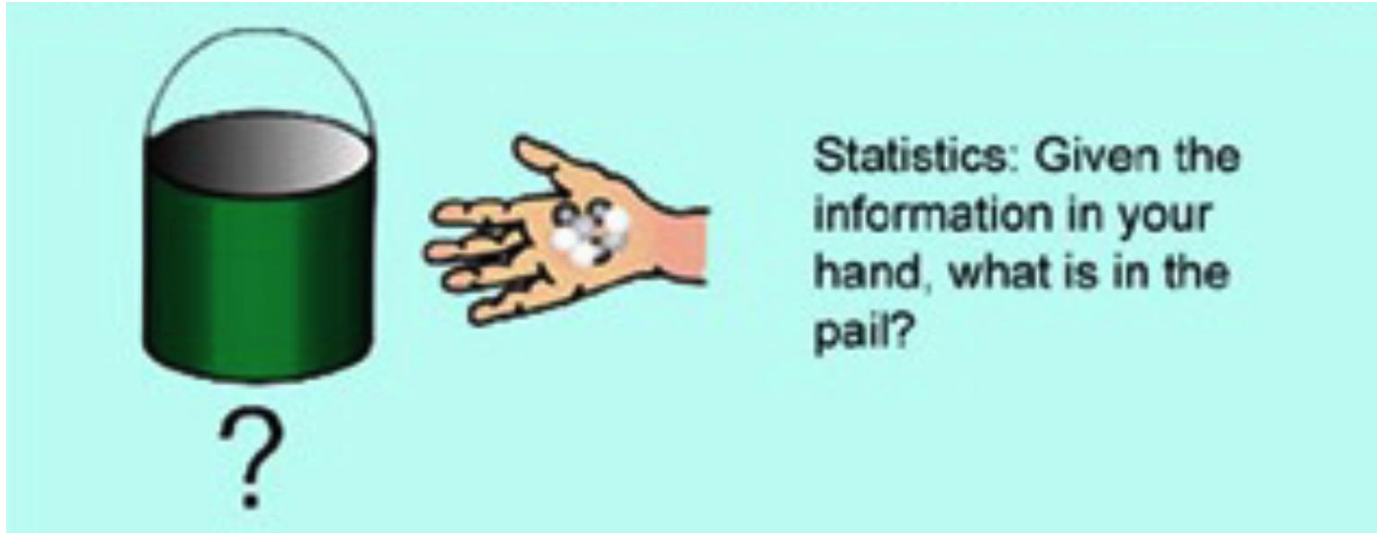
Probability

- A formality to make sense of the world in terms of uncertainty.
 - Numerical descriptions of how likely an [event](#) is to occur



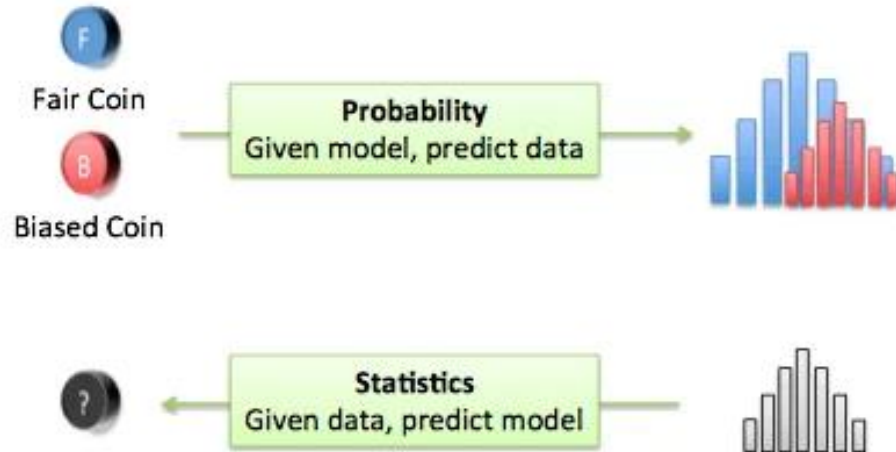
Statistics

- The discipline that extracts correct information from data



Probability and Statistics

Probability & Statistics



Summary: Probability and Statistics

- **Probability** is about determining how likely different outcomes are.
- **Statistics** is about using observed data to understand the process that generated it and estimating the parameters of the model.

Blind Box Example

- If the seller told the truth, the probability that all ten of you got a Voldemort is < 0.001 . —→ Probability
- Given the above observation, you are pretty confident that the seller was lying. What's the true chance of getting a Harry Potter versus Voldemort? —→ Statistics
- How should the seller set the distribution of items in the boxes to balance the rarity and desirability of items, thereby increasing sales? —→ Optimization
- Based on historical data, how should the seller improve his/her prediction of customer preferences and future sales trends? —→ Machine learning

Questions

1. Given a dataset, you want to predict how the data is generated. Which theory you may use?
 - a) Probability
 - b) Statistics

2. Given the profit as a function of strategies, you want to maximize the profit. Which theory you may use?
 - a) Simulation
 - b) Optimization

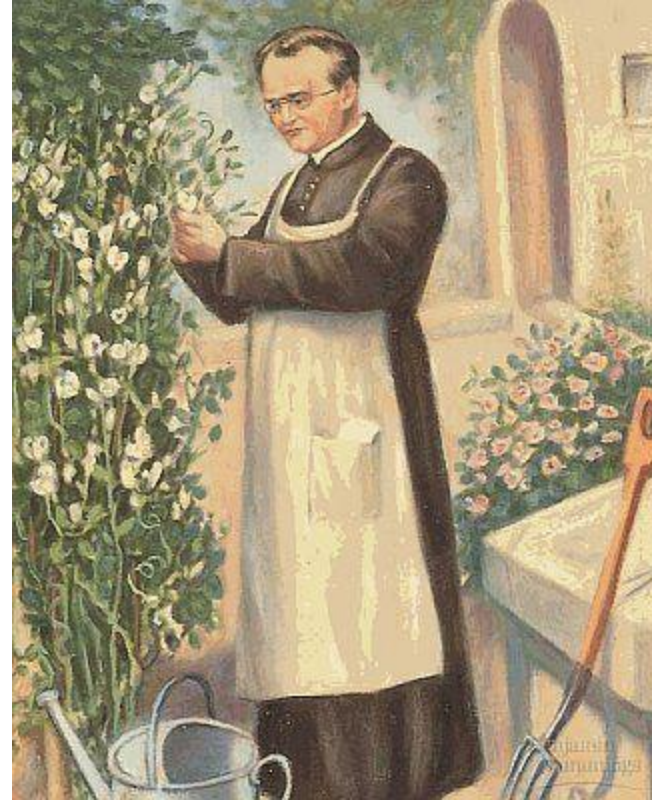
Questions

1. Given a dataset, you want to predict how the data is generated. Which theory you may use?
 - a) Probability
 - b) **Statistics**

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 - b) **Optimization**

Example 2

The Victory of Data Science over Human Imagination

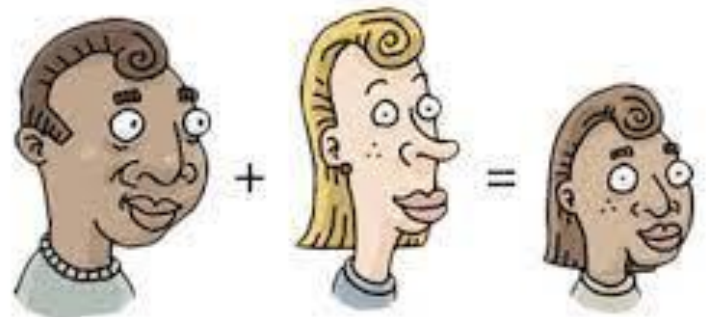


Setting the Stage: The Early Misconceptions of Heredity

Before data science, humans relied on imagination and assumptions to explain heredity. One popular belief was **blending inheritance**:

The Belief in Blending Inheritance:

- People thought traits of offspring were an *average* of their parents:
 - *Red + White = Pink*
 - *White + Black = Grey*



Why This Idea Was Problematic:

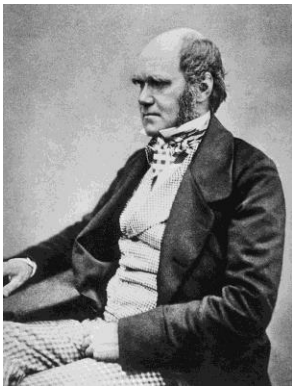
- If blending inheritance were true, beneficial traits (like strength, speed, or intelligence) would mix and weaken over generations.
- This would make evolution through natural selection impossible, as new advantageous traits could never become dominant.

Darwin's Struggle:

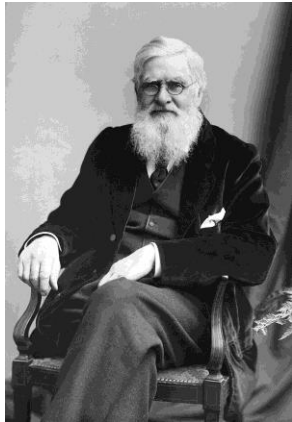
- Charles Darwin's theory of evolution relied on the idea that beneficial traits could accumulate over time.
- But he couldn't explain how traits were inherited.

The Issue:

- Darwin did **not collect enough data**.
- He also did not **analyze the data** he had deeply enough to develop a new theory.



Charles Darwin



Alfred Wallace

A letter from Charles Darwin to Alfred Wallace

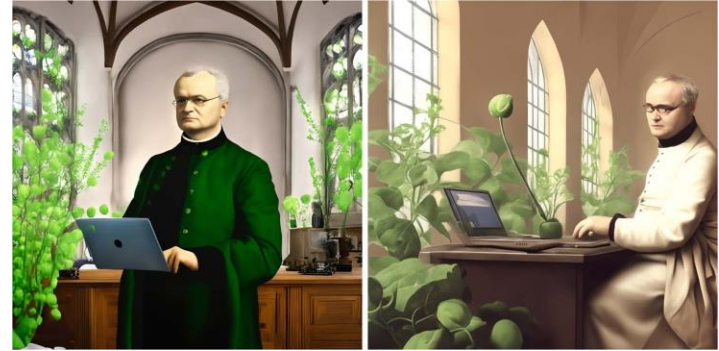
"I do not think you understand what I mean by the non-blending of certain varieties. It does not refer to fertility; an instance I will explain. I crossed the Painted Lady and Purple sweetpeas, which are very differently coloured varieties, and got, even out of the same pod, both varieties perfect but not intermediate. Something of this kind I should think must occur at least with your butterflies & the three forms of Lythrum; tho' those cases are in appearance so wonderful. I do not know that they are really more so than every female in the world producing distinct male and female offspring."

Takeaway:

Imagination alone wasn't enough. A data-driven approach was needed to solve this mystery.

Mendel's Revolutionary Discovery: Data > Assumptions

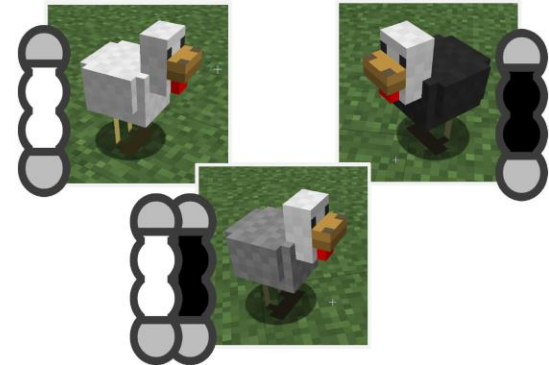
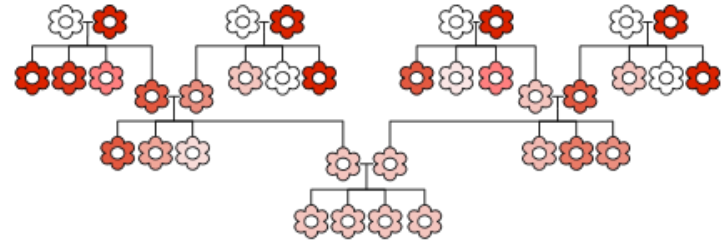
Gregor Mendel, the father of modern genetics. Unlike his predecessors, Mendel focused on **data collection** and **scientific experimentation** to uncover how traits are inherited.



Mendel's Experiments: The Power of Observation

What Did People Believe?

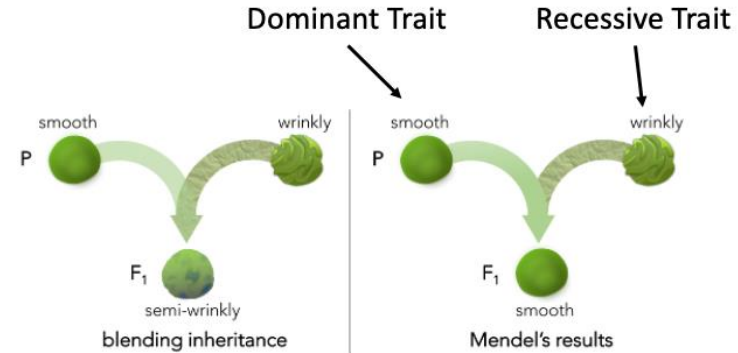
- Before Mendel, most people believed offspring traits were a blend of their parents' traits.
- For example:
 - A purebred tall plant crossed with a purebred short plant would produce medium-sized plants.



Mendel's Experiments: The Power of Observation

What Did Mendel Observe?

- Mendel conducted controlled experiments on pea plants.
- He cross-pollinated purebred plants with specific traits (e.g., smooth seeds vs. wrinkled seeds).
- The results were shocking:
 - The offspring resembled **one parent**, not a blend of both:
 - *Smooth seeds + Wrinkled seeds = Smooth seeds.*
 - This led Mendel to discover **dominant** and **recessive** traits.



Mendel's Principles of Inheritance

After years of **meticulous experimentation**, Mendel proposed three fundamental principles:

Principle of Uniformity:

When two purebred parents with contrasting traits are crossed, the offspring will resemble only the dominant parent (e.g., Smooth dominates Wrinkled).

Principle of Segregation:

Traits don't disappear; recessive traits can reappear in later generations.

For example:

In the second generation, some offspring of *Smooth + Wrinkled* could still have wrinkled seeds.

Principle of Independent Assortment:

Traits are inherited independently of one another.

For example:

Whether a plant has smooth seeds doesn't affect its flower color.

The Importance of Large Data

Mendel spent **eight years** collecting data from thousands of pea plants.

- Why was this important?
 - A **large sample size** ensured the patterns he observed were statistically valid and not just random chance.
 - Mendel's approach highlights the importance of **data-driven conclusions**.

Mendel's Use of Probability and Statistics

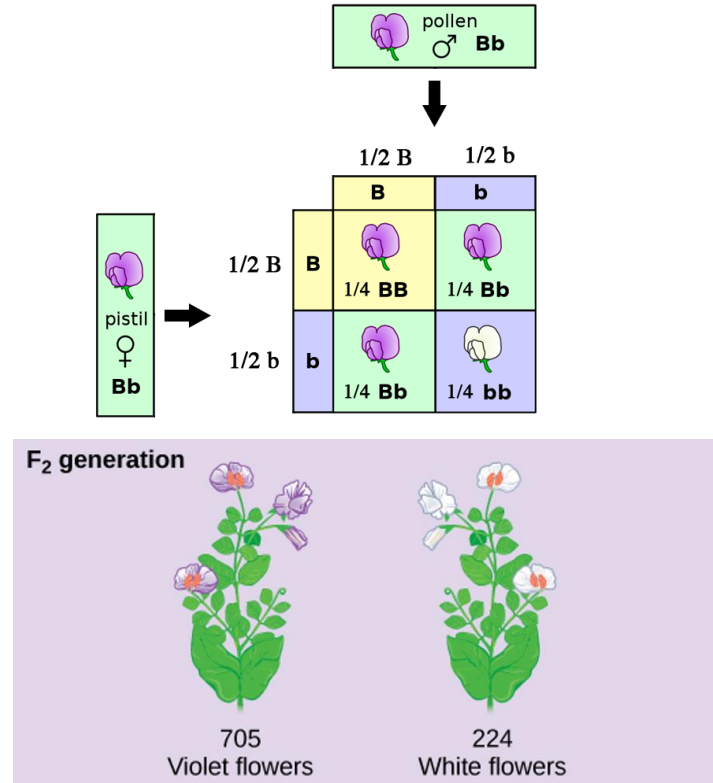
Mendel didn't just collect data—he used **data science tools** to test and validate his findings.

Testing Hypotheses with Probability

Mendel created a probability model to predict how traits would be passed on.

- **Example:**

- Mendel predicted that 75% of the descendants of certain pea plant crosses would exhibit the dominant trait.
- **Result:**
 - When he observed the data, 75.9% of the descendants showed the dominant trait—matching his prediction almost perfectly.

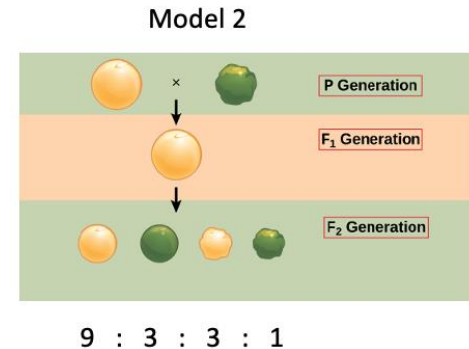
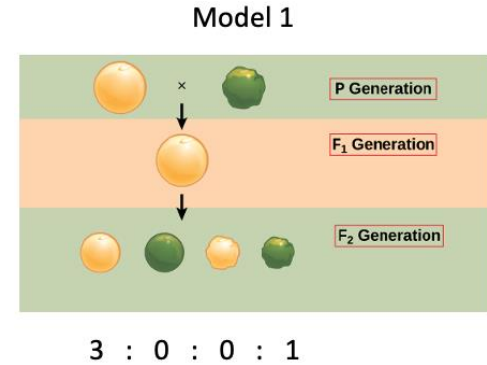


Using Models to Test Competing Theories

Mendel also tested whether traits were dependent or independent of each other.

- **The Experiment:**

- He crossed plants hybrid for two traits (e.g., seed shape and seed color).
- Two competing models were tested:
 - **Model 1:** The traits are dependent on each other.
 - **Model 2:** The traits are independent.



Using Models to Test Competing Theories

Results:

The offspring followed a **9:3:3:1 ratio**:

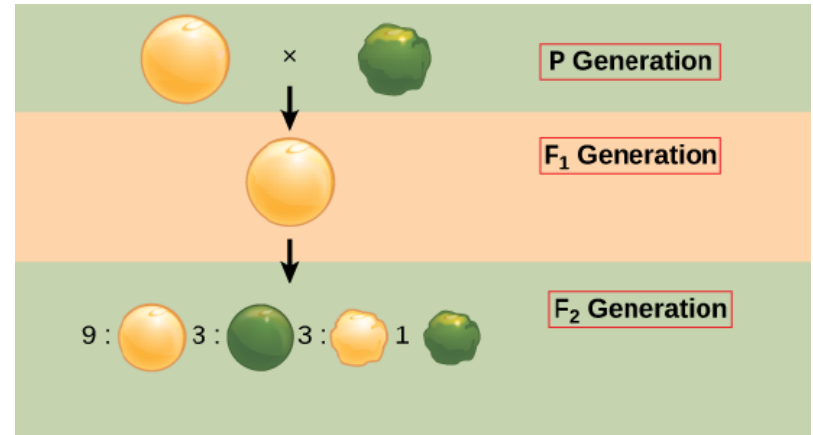
315 plants with round, yellow seeds.

108 plants with round, green seeds.

101 plants with wrinkled, yellow seeds.

32 plants with wrinkled, green seeds.

This confirmed that the traits were **independent**.



Takeaway:

Mendel's ability to test models with real data laid the foundation for modern genetics.

The Role of Modern Data Science

Today, we have tools that Mendel could never have imagined. These tools enable us to make discoveries faster and on a much larger scale.

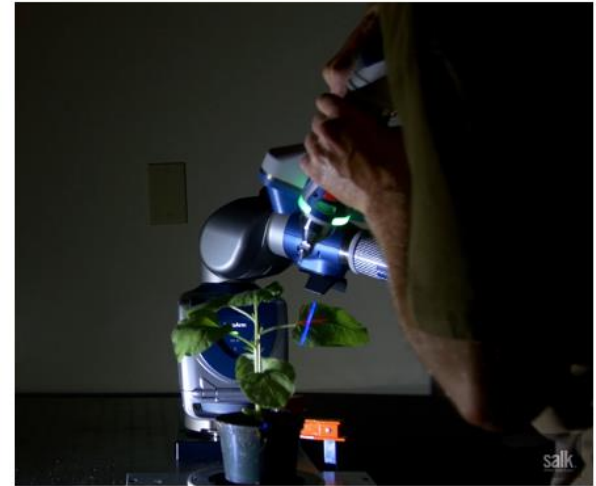
Automation and Machine Learning:

In Mendel's Time:

- Mendel manually observed and recorded traits by hand—a slow and tedious process.

Today:

- Scientists use automated camera systems to track plant traits across thousands of samples in real-time.
- Machine learning algorithms analyze complex 3D shapes of branches, leaves, and seeds to uncover patterns Mendel couldn't observe.



A Salk technician 3D scanning a plant

Applications of Mendel's Principles:

Plant Breeding:

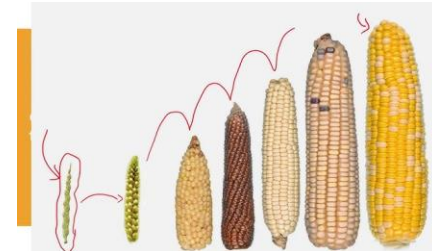
Mendel's principles help scientists develop new crop varieties that are more resistant to disease, drought, and pests.

Animal Breeding:

Genetic optimization is used to improve livestock traits, such as milk production or growth rate.

Medicine:

Mendel's work paved the way for understanding how diseases are inherited and developing personalized treatments based on genetics.



Why Mendel's Story Inspires Data Scientists

Mendel's work teaches us several important lessons:

- **Start with Data:**

Instead of relying on assumptions, Mendel let the data guide his discoveries.

- **Use Models to Make Predictions:**

Mendel used probability models to predict outcomes and test hypotheses.

- **Iterate and Improve:**

Mendel spent years refining his experiments and analyzing his data.

The Optimal Garden Layout: A Fun Challenge

Imagine you're a gardener with a passion for creating the perfect flower bed. But this isn't just any garden—you want to **strategically design your garden layout to create new flower colors!**



The Rules of the Garden Game (Constraints)

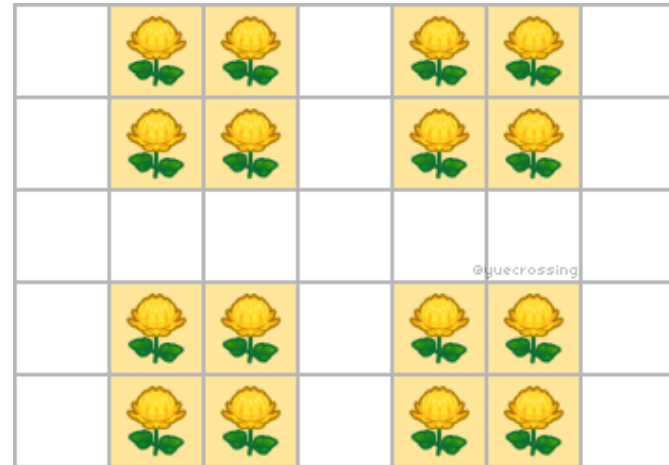
- Flowers have a chance to reproduce once per day
- If a flower rolls "successful" at reproduction, it will then look for a valid partner (same species) in the 3x3 space around itself to breed with, and then produce offspring.
- If it does not find a valid partner, it will make a cloned offspring
- Breeding takes precedence over cloning whenever possible
- When a flower is successful, it (and its partner) are marked as invalid or "locked" and cannot be produced until the next day



Your Goal: Create Violet Flowers!

You have a 5-by-7 garden, and you've planted **yellow flowers**.

- **Objective:** Use your yellow flowers to create **violet flowers** through breeding!
- **Challenge:** Find the **optimal layout** of yellow flowers to maximize the production of violet flowers.



Why is This Problem So Tricky?

It's Complex!

Breeding and cloning introduce multiple layers of interactions.

Flower placement influences outcomes due to the 3x3 reproduction rule.

It's an Integer Programming Problem!

Integer programming is a type of optimization problem that's extremely difficult to solve directly.

It involves testing many combinations of flower placements to find the best layout—a task that quickly becomes overwhelming for large gardens.

The Bigger Picture: Optimization and Machine Learning

This fun garden problem is more than just flowers—it introduces powerful real-world concepts!

Optimization Algorithms:

In courses like **MAT3007**, you'll learn algorithms to solve problems like this, where resources (like flower placement) need to be optimized for the best results.

Machine Learning and Neural Networks:

Instead of solving the problem manually, you could train a machine learning model to predict the best garden layout!

Courses like **DDA3020** and **DDA4210** teach you how to use deep learning to tackle complex problems like this.

Example 3

How Data Science Tools Power Smarter Learning Apps

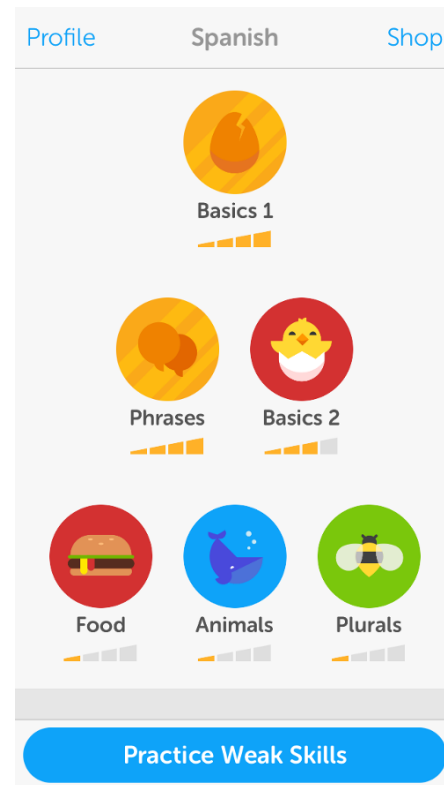
Creating an app to enhance student learning is an exciting challenge, but the real magic lies in **how data science tools make this possible.**

Let's break it down and explore why data science tools are so powerful.

Enhancing Student Learning Outcomes

Goal: Create a mobile app that enhances language learning for students—making the process *personalized, fast, enjoyable!*

Any ideas?



How Data Science Tools Come Together

Data Collection and Cleaning

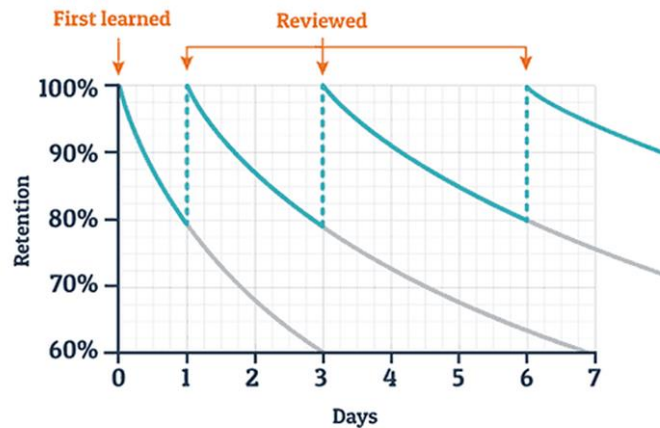
Collect data on review times, user responses, and word difficulties to create a clean, structured dataset.

Data Visualization

Plotting the **forgetting curve** to see how memory decays over time.

Visualizing how the number of repetitions affects memory retention.

Typical Forgetting Curve for Newly Learned Information



How to model the forgetting curve?

We can use, for example, an exponential model to describe the **probability** of recalling a word:

$$P\{q(t) = 1\} = \exp(-r(t) \cdot (t - t_{last}))$$

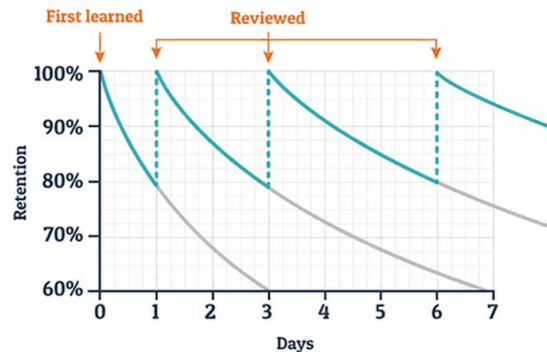
where:

- $q(t)$: a **binary** variable indicating if the user recall the word at time t
- t_{last} : the time of last review
- $r(t) \geq 0$: forgetting rate, which itself depends on many factors. For example,

$$r(t) = r_0 - a \cdot n(t)$$

- Here $n(t)$ represents the number of repetitions, and r_0, a vary among users

Typical Forgetting Curve for Newly Learned Information



How to learn the forgetting curve from data?

- Using tools from **statistics**, we can estimate r_0 and a !

What if the forgetting curve is more complex?

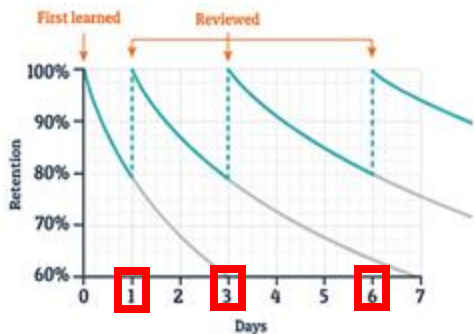
- Recall

$$\mathbb{P}\{q(t) = 1\} = \exp(-r(t) \cdot (t - t_{last}))$$

- The forgetting rate function $r(t)$ can depend on lots of factors with complicated probability models:
 - Language: English, French, Spanish,...
 - Words' difficulty level: CET4/6, TOEFL/IELTS, GRE/GMAT/LSAT,...
 - Time for reviewing: early morning, late night,...
 - User background: students, working class, children,...
 -
- With modern **machine learning**, we can account for all these factors and learn complex functional relationships.

How do we find the *optimal* time for reviewing?

- We observe that reviewing material at intervals over time enhances long-term memory



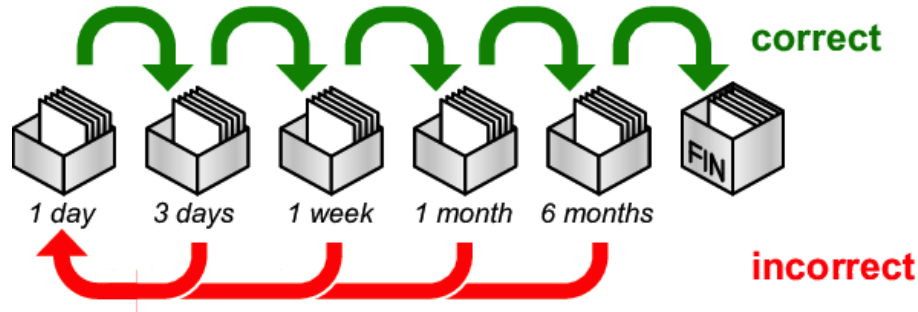
- “Fast”: To avoid forgetting, we want to flatten the curve as much as we can → increase the number of repetitions
- “Enjoyable”: the user would be painful if repeating the same task too many times → decrease the number of repetitions

How do we find the *optimal* time for reviewing?

- We need an **optimization** model!
- *Decision*: the time points that remind the user to review
- *Objective*: Maximizes the probability of memorizing words while minimizes pain of repetition
- *Constraints*: Duration of study, maximum number of repetitions, etc.
- Solving this mathematical optimization yields the optimal decision!

Learning and decision-making on the fly

- What if we do not have data for a new user?



Machine learning tools (Reinforcement learning) design “smart” interactions between the APP and the user (like the strategy shown in the above figure), so that it can jointly learn the user’s forgetting curve while optimizing the learning outcomes

Summary of this example

